

Aniket Mishra

Data Scientist & Data Engineer

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Professional Summary

Data Scientist and Data Engineer with 5 years of experience in designing **ETL pipelines, predictive models, and anomaly detection systems** for **10GW+ renewable energy assets**. Proven track record in reducing cloud costs by **55%**, accelerating analytics pipelines by **16×**, and capturing 98% of false and missing alarms. Published researcher in **Springer LNCS (IDA 2026)** on interpretable machine learning and subgroup discovery. Experienced in building **agentic AI systems**, with skills in Python, AWS, and advanced machine learning across both **academic research** and **industry-scale AI systems**.

Education

Eindhoven University of Technology (TU/e), Eindhoven, NL Sep 2024 - Aug 2026 (Expected)
MSc, Data Science and Artificial Intelligence

- **Majors:** Data Mining & Machine Learning; Data Engineering & Management
- **Minor:** Process Mining & Visual Analytics
- **Relevant coursework:** Machine Learning Engineering, Deep Learning, Process Mining, Visual Analytics, Survival Analysis, Foundations of Artificial Intelligence, Machine Learning for Industry, Big Data Management, Knowledge Engineering, Research Topics in Data Mining, Philosophy and Ethics of AI, Engineering Data Intensive Systems

SRM Institute of Science and Technology, Chennai, IN Jun 2016 - Sep 2020
B.Tech, Computer Science (*First Class with Distinction*)

- **Thesis:** *Image Super-Resolution using Feedback Loops in GANs*
- **Best Research Paper Award:** RCICD Conference 2020
- **Relevant coursework:** Python Programming, Artificial Intelligence, Database Management Systems, Pattern Recognition & Expert Systems in Medicine, Neuro-Fuzzy & Genetic Programming, Data Science & Big Data Analytics, Enterprise Resource Planning, Marketing Management, Digital System Design, Object-Oriented Analysis & Design, Algorithm Design & Analysis, Software Engineering Principles

Professional Experience

Eindhoven University of Technology (TU/e), Eindhoven, NL Feb 2026 - Aug 2026 (Expected)
Student Assistant

- Teaching Assistant for MSc courses: Machine Learning Engineering (2AMM15, Y2Q3) and Reinforcement Learning in Practice (2AMC15, Y2Q4).
- Maintain the RL course repository; implement new agent environments and CLI/UI features.
- Develop and maintain components of an automated grading system for programming assignments at the Advanced Models through Open Research and Engineering Lab.
- Mentor students through challenge-based learning, including regular feedback sessions.

Avathon (formerly SparkCognition), Bengaluru, IN Mar 2021 - Jul 2025
Data Scientist / Data Engineer

- Scaled real-time ETL pipelines for **10GW+** renewable assets, reducing API latency, onboarding **40+ assets**, and cutting cloud costs by **55%**.
- Built anomaly detection system configurable by asset/rule, **reducing false alarms by 98%** and **capturing 92%** of missed alarms.

- Developed predictive downtime models (2–4 months in advance) with **>95% accuracy**.
- Authored Python ETL framework adopted company-wide; **cut customer onboarding from 3 months to 1 day**.
- Optimised Elasticsearch queries from **4+ hours to ~15 minutes per asset**.
- Optimized AWS Lambda and containers, lowering compute and storage costs by **48%**; reduced site data model memory usage by **50%**.
- Deployed Streamlit apps for KPI validation, hyperparameter tuning, and monitoring; **used by 20+ stakeholders**.
- Mentored and onboarded **5+ engineers**, improving project delivery timelines.
- Improved MCR processing accuracy to **99% OEM parity**.

Infosys Ltd., Mysore/Hyderabad, IN

Jan 2020 - Mar 2021

Systems Engineer, Data & Analytics (incl. Internship)

- Internship top performer; completed **2 capstone projects** (Banking & Insurance).
- Managed **2000+ ETL workflows** in Informatica PowerCenter with **99.9% uptime**.
- Executed database migrations with **zero data loss**.

Data Science Instructor (Freelance), Remote

Aug 2023 - Sep 2024

- Delivered corporate training in analytics, ML, and engineering to professionals.
- Mentored learners into analytics roles with portfolio projects and interview prep.

Phoenix Software Solution, Bhubaneswar, IN

May 2018 - Jun 2018

Web Developer Intern

- Developed GST billing and tax-invoice web platform using PHP/MySQL.
- Built responsive frontend using Bootstrap, JavaScript, and AJAX.

Research & Publications

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- **Published:** International Symposium on Intelligent Data Analysis (Springer LNCS, vol. 16513, IDA 2026). Mishra, A., Cărbunaru, C., Duivesteijn, W., “**Exceptional Model Residual Mining, and Three Richer EMM Description Languages**,” 2026. Paper | Code
 - **Best Paper Award:** Research Conference on IoT, Cloud & Data Science (RCICD 2020). Mishra, A. *et al.*, “**SRFBGAN: Image Super-Resolution using Feedback Loops in GANs**,” *JETIR*, 2020. Link

Conferences & Hackathons

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- **IDA 2026, Leiden** (Apr 2026): Presented paper on Exceptional Model Residual Mining. Three days of talks on optimization, explainable ML, and intelligent data analysis.
 - **bunq 7.0 Hackathon** (Apr 2026): Built ShieldPay, an agentic fraud prevention system using Claude tool use, bunq API, and FastAPI. 24-hour build with one teammate.

Projects

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- **ShieldPay** FastAPI, Claude (tool use), bunq API
Agentic payment fraud prevention system. An LLM agent assesses merchant risk from checkout screenshots and online reviews, checks transaction history, and generates disposable virtual cards via the bunq API. Frontend streams the agent’s reasoning live. No hardcoded merchant lists; works globally from day one. Devpost | Repo
 - **Explainable Movie Recommendation System** Dash, XGBoost
A real-time movie recommendation system using clustering, PCA, and live dashboards with interpretable explanations for each recommendation. Repo
 - **Competitive Sudoku AI** Monte Carlo, MCTS, Heuristics
AI agents built for competitive Sudoku. Benchmarked against multiple algorithms including Monte

Carlo Tree Search and Markov Chains. Repo

- **Third Eye – Threat Detection (Kerala Police Prototype)** OpenCV, Jetson Nano
Real-time facial recognition prototype integrated with police DB; deployed on edge hardware. Repo
- **Reinforcement Learning Robot Navigation** PyTorch (PPO, DQN)
Implemented multiple RL algorithms for autonomous robot delivery; PPO outperformed others. Repo
- **Historical Shark Attacks Visualization** Dash/Plotly
Interactive dashboard for historic Australian shark attacks. Repo
- **Heart Failure Severity Detection** Keras, Healthcare AI
NN classifier (levels 1–4) built with doctors at AIIMS. Repo
- **dotfiles: Dev Productivity Toolkit** Rust, Zsh, Bash
Automations for macOS/Linux (CLI tools, shell functions, fast setup scripts). Repo
- **scripts: Automation Utilities** Python, Rust, OpenCV
Small tools for everyday workflows: PDF/image ops, batch file tasks, and CLI helpers. Repo
- **statistical-model-implementer** Python, ML
Prototype library to auto-apply/evaluate models across datasets, compare metrics & models. Repo
- **quickSilver: Graph Path Query Estimator & Engine** C++17, CMake
Designed and implemented a cardinality estimator and fast path query evaluation for a C++ in-memory graph database (directed, labeled RDF-style graphs & simple path queries). Project

Full archive available on *GitHub* and *Portfolio Website*.

Leadership & Volunteering

- **Codezoned (NPO), Core Team & Developer (2018–Present):** Moderated multiple large open-source repositories (incl. *ScriptsDump*, a major CS/Math script archive); managed community servers and maintained the organization’s website.
- **Hack Against Covid 2020 (MLH, Devfolio):** Technical Team Lead: Led infra automation, registration system, and community management for **1000+ participants** from **15+ countries**.
- **ALERT (Alertathon):** Organizing Volunteer for the event “Run to save a life,” Chennai (2018).

Awards & Competitions

- **Best Research Paper Award**, RCICD 2020.
- **Winner (1st)**, IIT Madras Robotics (Arduino), 2018.
- **Top Qualifiers**, HAC-KP 2020 Hackathon.
- **bunq 7.0 Hackathon**, Built ShieldPay (agentic fraud prevention), Apr 2026.

Skills

- **Programming:** Python, SQL, Rust, JavaScript, C++, Bash
- **ML/AI:** PyTorch, Transformers, AutoML, Anomaly Detection, Time Series Forecasting, Reinforcement Learning, GNNs, LLMs, AI Agents, Generative AI, Stable Diffusion
- **Data Engineering & MLOps:** Airflow, Docker, Serverless, CI/CD, Terraform
- **Databases & Search:** PostgreSQL, MySQL, InfluxDB, Elasticsearch, DuckDB, Neo4j, ChromaDB
- **Cloud:** AWS, GCP, Heroku, Netlify
- **Visualization:** Plotly, Dash, Streamlit, Panel
- **Tools & Frameworks:** FastAPI, REST APIs, Git, Swagger, Postman, Jira, Confluence, Ollama, ComfyUI, Poetry, UV, Warp

Certifications

- **Statistics with Python Specialization** (Coursera), 2020
- **Programming, Data Structures & Algorithms with Python** (NPTEL), 2019

- **Python Bootcamp: Zero to Hero** (Udemy), 2018

Additional Information

- **Languages:** English (Fluent), Hindi (Fluent), Odia (Native), German (Beginner), Japanese (Beginner)
- **Work Eligibility:** EU student residence permit
- **Interests:** Research in LLMs, AutoML, and data-intensive systems; community hackathons and open-source projects
- **Teaching:** Delivered corporate training on ML & Data Engineering; mentored professionals into analytics roles
- **Hackathons:** Participant and mentor at 5+ hackathons; winner at IIT Madras Robotics, top qualifier HAC-KP
- **Open Source:** Contributor and maintainer on multiple GitHub projects
- **Creative Interests:** Writing poems; using storytelling and pictures to explain concepts

References available on request.